

Graphical System Design Accelerates System Development

Using graphical system design, engineers can invest in a single development platform from which multiple solutions can be derived to serve a broad range of application needs



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Successful product-oriented companies choose a vector of differentiation to deliver to their customers: best in class, fastest time to market and lowest cost. In the global arena, companies are encountering nimble competitors able to compete on multiple fronts simultaneously. To get ahead of these disruptive forces, engineers must shed ideological thinking and focus on pragmatic outcomes using whatever tools necessary to ensure the fastest and most accurate results.

However, evaluating tools and technology comes at a cost; let alone learning and implementing them. This initial investment can create a barrier strong enough to prevent engineers from using the technology needed to achieve a competitive advantage.

Abstraction of commercial technologies

A graphical system design platform can abstract the complexity of field-

programmable gate array (FPGA) technology to the pin. A clear graphical representation of system functionality replaces thousands of lines of equivalent VHDL code. The same platform also abstracts programming complexity when taking advantage of multicore processors or digital signal processors (DSPs).

Any other commercial standard technology, including communication technologies and protocols, is abstracted in the same manner. With this abstraction of technology complexity, engineers can focus on using technology to serve the end goal of the system. Without this approach, engineers seeking competitive advantage through better performance and lower costs of commercial technology either have to focus on programming components or interface with specialists who can do so. Both options increase time to market.

On the other hand, graphical system design accelerates development by sufficiently abstracting complexity yet providing access to the pin, making it possible for engineers to easily use technology for the purpose of the system. With the graphical system design approach, engineers can gain the performance and cost benefits of commercial technology without increasing time to market.

Integration of multiple software methods

Graphical system design enables engineers to explore multiple ways of solving a problem and lock in on the best options more quickly than traditional methods. As engineers visualise system functionality, different system components might need different methods or models of computation to best describe that functionality. For example, parallel program-

ming is best represented as a graphic, but equations are represented well using text.

The structure of the system may be state-based, sequential, parallel based on data flow or a mixed model (refer Fig. 1). Graphical system design software can incorporate multiple models or methods of computation within a single exploration space to give engineers access to the best methods for the functionality they need. In doing so, graphical system design abstracts complexity at the system level, where system components of different models can be placed together in a single software platform space, and then integrated together visually, functionally and in an architecturally sound way.

Customisable off-the-shelf

The graphical system design approach includes both software and hardware as part of the exploration and implementation platform. Often engineers have excellent high-level software tools, but when they get to the real implementation of prototypes or end systems, the tool chain starts slowing down development.

With graphical system design, the single open-platform approach integrates software with customisable off-the-shelf hardware platforms from beginning to end. This approach takes a comprehensive view of the system for the end purpose of functioning as a controller for wind turbines, an automated test system for cellphones, or a robot that can perform surgery on humans.

Every system that needs measurement and control requires hardware implementation. Graphical system design includes the same platform elements in multiple hardware form factors, making it possible for engineers with customis-